MIL-C-15734B(SH)
3 June 1976
SUPERSEDING
MIL-C-15734A(SHIPS)
2 October 1967
(See 6.6)

MILITARY SPECIFICATION

COIL ASSEMBLY, COMPASS COMPENSATING

(DEGAUSSING), TYPE "R-2"

‡ This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1. This specification covers type "R-2" compass compensating coils which are used to neutralize the magnetic field due to degaussing in the vicinity of a magnetic compass.

2. APPLICABLE DOCUMENTS

‡ 2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

FF-W-92 - Washer, Flat (Plain).
QQ-B-613 - Brass, Leaded and Non-Leaded; Flat Products (Plate, Bar, Sheet and Strip).
QQ-B-626 - Brass, Leaded and Non-leaded: Rod, Shaped, Forgings, and Flat Products With Finished Edges (Bar and Strip).
QQ-B-650 - Brazing Alloys, Copper, Copper-Zinc and Copper-Phosphorus.
QQ-S-571 - Solder; Tin Alloy; Lead-Tin Alloy; and Lead Alloy.
TT-L-58 - Laquer; Spraying, Clear and Pigmented, (General Interior Use).
TT-P-645 - Primer, Paint, Zinc-Chromate, Alkyd Type.

MILITARY

MIL-W-583 - Wire, Magnet, Electrical.
MIL-I-631 - Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-E-699 - Enamel, Deck, Gray, Exterior (Formula No. 20).
MIL-R-900 - Rubber, Gasket Material, 45 Durometer Hardness for Moderately Low Temperature Service.
MIL-S-901 - Shock Tests, H. I. (High-Impact): Shipboard Machinery, Equipment and Systems, Requirements for.
MIL-C-915/14 - Cable, Electrical, 600 Volts, Type MHOF.
MIL-E-917 - Electric Power Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-P-997 - Plastic Material, Laminated, Thermosetting, Electrical Insulation: Sheets, Glass Cloth Silicone Resin.
MIL-P-15024 - Plates, Tags and Bands For Identification of Equipment.
MIL-P-15024/5 - Plates, Identification.
MIL-P-15037 - Plastic Sheet, Laminated, Thermosetting, Glass-Cloth, Melamine-Resin.
MIL-I-16923 - Insulating Compound, Electrical, Embedding.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts; Packaging and Packing of.
MIL-C-20079 - Cloth, Glass: Tape, Textile Glass; and Thread, Glass.
MIL-I-24092 - Insulating Varnish, Electrical, Impregnating, Solvent Containing.

STANDARDS

MILITARY

MIL-STD-167-1 - Mechanical Vibrations of Shipboard Equipment.
MIL-STD-681 - Identification Coding and Application of Hook Up and Lead Wire.

FSC 5950

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.

DRAWING
NAVSHIPS 803-4759446 - Compass Compensating Coil Type "R-2".

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Sample for first article inspection. Prior to beginning production, a sample shall be tested as specified in 4.3 (see 6.4).

3.2 General requirements. The "R-2" compass compensating coil assembly shall conform to MIL-E-917 and Drawing 803-4759446 except as otherwise specified herein. In case of conflict between information shown on the drawing and this specification, this specification shall govern.

3.3 Material. Materials of construction shall be nonmagnetic and entirely suited for the intended application of mounting in the weather aboard ship.

3.3.1 Plastic. Plastic materials shall conform to MIL-P-997 and MIL-P-15037.

3.3.2 Magnet wire. Magnet wire shall conform to MIL-W-583.

3.3.3 Fibrous glass cloth. Fibrous glass cloth, tape, and thread shall conform to MIL-C-20079.

3.3.4 Brass. Brass components shall conform to QQ-B-626, FF-W-92, and QQ-B-613, unless otherwise specifically approved by the command or agency concerned.

3.3.5 Synthetic rubber gaskets. Synthetic rubber gasket materials shall conform to MIL-R-900.

3.3.6 Soft solder. Soft solder shall conform to QQ-S-571.

3.3.7 Brazing solder. Brazing solder shall conform to QQ-B-650.

3.3.8 Lacquer. Lacquer shall conform to TT-L-58.

3.3.9 Paint. Paint shall conform to MIL-E-699.

3.3.10 Paint primer. Paint primer shall conform to TT-P-645.

3.3.11 Insulating varnish. Insulating varnish shall conform to MIL-I-24092.

3.3.12 Insulating materials. All insulating materials shall be class A of MIL-E-917.

3.4 Insulation. All assemblies shall be insulated for operation as specified herein in a moisture-laden atmosphere. Every precaution shall be taken against the possible failure of the insulation due to a combination of heat and moisture. All insulated electrical windings shall be thoroughly impregnated by the vacuum-pressure process which will insure thorough penetration and a complete filling of all interstices with a suitable protective varnish or compound.

3.5 Components. The type "R-2" compass compensating coil assembly shall consist of five separate coils rigidly arranged in the configuration shown on Drawing 803-4759446. The ends of the various coil windings shall terminate in terminal boxes where they shall be connected to the individual conductors of two type MHOE-14 cables in accordance with MIL-C-915/14 by means of pressure-type, nonmegnetic connectors. The entire coil assembly shall be embedded in an embedding insulating compound conforming to MIL-I-16923 in such a manner as to fill all void spaces in the enclosure. Details of the coil assemblies are indicated in tables I and II and shown on Drawing 803-4759446. The individual windings of the coils shall be insulated from each other and the coil assembly thoroughly impregnated by a method which will ensure the evacuation of all air and water from, and the filling in of, all the interstices within the coil windings prior to installation of the coils in the enclosure.

Source: https://assist.dla.mil -- Downloaded: 202640-18T07:33Z
Check the source to verify that this is the current version before use.

Table I - Coil assembly characteristics.

	Heeling coil	NE/SW coils	NW/SE coils
No. coils	1	2	2
No. windings per coil	4	4	4
No. turns - outerwinding #1	153	365	365
No. turns - winding #2	102	360	360
No. turns - winding #3	51	342	342
No. turns - inner winding #4	51	342	342
Wire size - AWG single paper covered Formvar or similar	#20(0.0355 inch)	#21(0.0319 inch)	#21(0.0319 inch)
Turns per layer-outer winding #1	51-3 layers	73-5 layers	73-5 layers
Turns per layer-outer winding #2	51-2 layers	90-4 layers	90-4 layers
Turns per layer-outer winding #3	51-1 layers	114-3 layers	114-3 layers
Turns per layer-inner winding #4	51-1 layers	114-3 layers	114-3 layers

Table II - Coil assembly dimensions.

	Heeling coil inches	NE/SW coils inches	NW/SE coils inches
Winding length (approx.) #1	2.000	2.620	2.620
Winding length (approx.) #2	2.000	3.380	3.380
Winding length (approx.) #3	2.000	4.000	4.000
Winding length (approx.) #4	2.000	4.000	4.000
Maximum outside diameter	9.532	2.125	2.125
Minimum inside diameter	-----	0.688	0.688
Outside diameter of enclosure - (max)	10.00		
Inside diameter of enclosure - (min)	4.312		

3.6 Coil assembly characteristics. The coil assembly shall have the characteristics indicated in table I.

3.7 Dimensions. Dimensions of the type "R-2" compass compensating (degaussing) coil assembly shall be in accordance with table II.

3.8 Enclosure. Approximate configuration and limiting dimensions of the case are shown on Drawing 803-4759446. Construction shall be such as to minimize weight without sacrificing strength. Soft soldered joints shall be a minimum, and where used shall act as a sealing agent only, not as a strength member. Where sheet brass is used for the enclosure, both surfaces of joints sealed by solder shall be first tinned or tin plated.

3.8.1 Watertight integrity. The enclosure and terminal box shall be capable of withstanding the watertight test specified in 4.5.3 and shall not be affected by salt water.

3.8.2 Mounting. The compass compensating coil mounting surfaces on the four mounting brackets shall be flush with the compass compensation coil cover.

3.8.3 Color. The enclosure shall be painted with gray paint as specified in 3.3.9.

3.9 Electrical connections. Two 20-foot lengths of type MHOF-14 cable in accordance with MIL-C-915/14 shall be used for the cable leads. The two leads shall be spaced 180 degrees apart, one centrally located between the NE and SE coils, the other centrally located between the NW and SW coils. The connections from the two cable leads may be made directly to the coils by pressure-type or other nonmagnetic connectors, providing the connectors and leads are secured so as to prevent vibration or displacement. Spare conductors shall be dead-ended in the terminal boxes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.

3.9.1 The left cable, as shown on Drawing 803-4759446 shall include leads from all No. 1 and No. 2 windings. The right cable shall include leads from all No. 3 and No. 4 windings. Each cable lead free end conductor shall be identified using a reusable plastic tag with black figures on a white background. For example, "H1" represents #1 winding of heeling coil, "NE3" represents #3 windings of the two coils that produce the NE/SW magnetic field and "NW4" represents #4 windings of the two coils that produce the NW/SE magnetic field. Corresponding numbered windings ofdiametrically opposite NE/SW coils shall be connected in series such that the magnetic field produced by the windings will be aiding when measured on the axis of the assembly 4 1/2 inches above the horizontal mounting plane (see 4.5.1). Opposite NW/SE windings shall similarly be connected in series aiding. Individual internal conductors shall be distinctly color coded in accordance with MIL-STD-681. Conductor ends shall be marked to identify associated windings through the use of sleeving. Sleeving shall be type F tubing, or equal, in accordance with MIL-I-631. Identification markings shall be permanently indented, not stamped, on the sleeve in a contrasting color.

3.10 Electrical requirements.

3.10.1 Magnetic strength, polarity and symmetry. The completed assembly shall pass the magnetic tests specified in 4.5.1.

3.10.2 Insulation resistance The insulation resistance, when measured as specified in 4.5.2, shall be not less than 150 megohms (MΩ).

3.10.3 Dielectric strength. There shall be no evidence of insulation breakdown after being subjected to the dielectric strength test specified in 4.5.4.

3.11 Shock. The completed assembly shall meet the requirements of MIL-S-901 for grade A, class I, type A equipments as specified in 4.5.5.

3.12 Vibration. The equipment shall be designed to meet the vibration requirements of MIL-STD-167-1, type I environmental vibration from four to 33 hertz (Hz) (see 4.5.6).

3.13 Identification plate. An identification plate, brass, conforming to MIL-P-15024 and MIL-P-15024/5 which contains the following information shall be riveted securely to the periphery of the enclosure on the outboard surface near the intercardinal coil labeled "NE".

(a) Manufacturer.
(b) Contract number.
(c) Type "R-2" Compass Compensating Coils.
(d) National stock number _____

No part of this plate or fastenings shall extend more than 1/16 inch beyond the enclosure.

3.14 Compensation instructions An instruction plate of material identical to that of identification plate shall be attached on the outboard surface near the intercardinal coil labeled "NW". Black letters approximately 1/16 inch high, etched or otherwise permanently marked in a color to contrast with the background will be satisfactory. The instruction plate shall include the following.

"COMPENSATION. On any heading, set the current in control resistors labeled 'H' at a value to cause dip needle at compass position to remain level when DG current is on or off. Then place ship's head on NE or SW by compass to set current in resistors labeled "NE/SW", or NW or SE by compass to set current in resistors labeled "NW/SE". Adjust control resistor until compass remains steady on initial reading when DG current is on or off. Repeat entire procedure for finer compensation.

"DO NOT EXCEED 1.0 ampere per winding."

3.15 Marker plate. A marker plate of material identical to that of the identification plate of 3.13 and manufactured in accordance with Drawing 803-4759446 shall be attached on the outboard forward surface of the coil assembly enclosure midway between the NE and NW coil locations and on the compass compensation coil assembly centerline.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 First article inspection. First article inspection shall consist of the examination of 4.4.1 and tests specified in 4.5.

4.4 Quality conformance inspection.

4.4.1 Examination. Each unit coil assembly shall be subjected to surface examination to determine conformance with this specification. Surface examination shall include visual examination for defects and finish, workmanship, and measurements for external dimensional limitations specified on Drawing 803-4759446. Any item which fails to conform to the requirements shall be rejected.

4.4.2 Quality conformance tests. Sample coils shall be subjected to the tests of 4.5 with the exception of shock and vibration (4.5.5 and 4.5.6).

4.4.3 Sampling. As coil assemblies are produced and the number of finished units increases, sample coil assemblies shall be selected for quality conformance tests, as shown in table III. Until 12 samples have passed all of the tests, no failures are allowed. If any failure occurs, the cause shall be found and a design change to correct the cause in all assemblies produced shall be accomplished. Sampling shall begin again at the first line of the table and progress downward. At no time shall the cumulated number of failures exceed the number allowed for the number of assemblies produced and the number of assemblies tested as shown in the table. If the cumulated number of failures equals the disapproval number, all test data shall be reported to the command or agency concerned for review.

Table III - Sampling for quality conformance tests.

Cumulated production coil assemblies	Cumulating sample to be tested	Approval number	Disapproval number
Up to 5	3	0	1
6 to 9	4	0	1
10 to 14	5	0	1
15 to 20	6	0	1
21 to 28	7	0	1
29 to 36	8	0	1
37 to 45	10	0	1
46 to 56	11	0	1
57 to 68	12	0	1
69 to 81	13	1	2
82 to 95	14	1	2
96 to 110	15	1	2
111 to 126	16	1	2
127 to 144	17	1	2
145 to 163	19	1	2
164 to 183	20	1	2
184 to 203	21	1	2
204 to 225	22	2	3
226 to 248	23	2	3
249 to 273	24	2	3
274 to 298	25	2	3

Source: https://assist.dla.mil -- Downloaded: 202610-18T07:33Z
Check the source to verify that this is the current version before use.

4.5 Test procedures.

4.5.1 Magnetic test. The completed coil assembly shall be tested for magnetic field strength, polarity, and symetry of the magnetic field produced by each winding. At a point of measurement on the axis of the assembly 4 1/2 inches above the horizontal mounting plane, established by the top surface of the assembly, the vertical field resulting from 1.0 ampere d.c. in any one winding of either the NW or the NE coil shall not exceed 5 millioersteds (less than 20 millioersteds for four NW or four NE coil windings connected series aiding). Similarly, the horizontal field resulting from 1.0 ampere d.c. in each winding of the NE coil (measured seperately) shall have a component no greater than 2 millioersteds in the NW direction (less than 8 millioersteds for four NE coil windings connected series aiding) nor shall the NW coil have a component greater than 2 millioersteds in the NE direction when 1.0 ampere d.c. flows in each winding of the NW coil (less than 8 millioersteds for four NW coil windings connected series aiding). At the same point, the horizontal field resulting from 1.0 ampere d.c. in each winding of the heeling coil shall not exceed 2 millioersteds (measured separately and less than 8 millioersteds for four heeling coil windings connected series aiding). The magnetic field direction, at the point of measurement, produced by current flowing from plus to minus in the NE, NW, and heeling coils shall be NE, NW, and vertically downward respectively.

4.5.2 Insulation resistance. The insulation resistance of all twelve circuits shall be checked between each other and between each circuit and the case, using a 600-volt megger or vibratester. Coils with readings less than 150 MΩ will be rejected. This test shall be performed before andafter the shock andvibration tests.

4.5.3 Watertight integrity. Assemblies shall be tested for watertight integrity of the enclosure, using external water pressure applied in a container fitted with stuffing tubes for entrance of cable leads. Insulation resistance measured between any two conductors of the cable lead shall be at least 50 MΩ after the assembly has been subjected to a pressure of 25 pounds per square inch (lb/in^2) for 5 minutes. Water with a wetting agent added shall be employed forthis test. No water shall have penetrated the enclosure during this period.

4.5.4 Dielectric strength. The assembly shall be submitted to a dielectric test voltage of 1500 volts between windings or between any winding and the case for a period of 1 minute.

4.5.5 Shock. Shock tests shall be conducted in accordance with MIL-S-901 (see 3.11). Failure which is defined as inability to perform principal functions shall result in rejection. Failure to perform principal functions is defined as follows:

(a) Breakage of any parts, including mounting bolts. Minor chipping or distortion of parts will be accepted when such chipping or distortion cannot in any way impair the operation or performance requirements of the equipment.
(b) Appreciable distortion of any parts beyond manufacturers' drawing tolerance, including the case and mounting appendages.
(c) Insulation resistance values lower than those permitted by this specification.
(d) Failure to pass the dielectric strength test.
(e) Failure to pass the magnetic test.

4.5.6 Vibration. The equipment shall be tested in accordance with MIL-STD-167-1, type I environmental vibration from four to 33 Hz. The unit shall be tested under normal loaded conditions to duplicate shipboard installation. Submission of a detailed test "setup" for Navy review shall be required prior to any testing. Test reports shall be submitted to the Naval Ship Engineering Center, Ship Safety, Noise/Vibration Damage Control Section and Magnetic Defense Section.

4.6 Test reports. The supplier shall prepare a first article test report in accordance with the data ordering document included in the contract or order (see 6.2.2). Results of all tests and detailed descriptions of any damage or malfunctions incurred, at what stage of the tests it occurred, as well as details of any corrective action, shall be furnished. The test report shall be a complete chronological record of all tests performed, and shall include all test data developed from electrical, magnetic, and shock and vibration tests. A copy of the first article test report shall also be furnished to the command or agency concerned.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.

4.7 Inspection of preparation for delivery. The preservation and packaging, packing, and marking shall be inspected for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

(Preparation for delivery requirements specified herein apply only to direct Government procurement. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.5.)

5.1 Preservation-packaging, packing and marking. The equipment, repair parts, and technical publications shall be preserved-packaged level A or C, packed level A, B, or C and marked in accordance with MIL-E-17775 (see 6.2.1).

6. NOTES

6.1 Intended use. This coil assembly is for use on U.S. Navy ships to produce a magnetic field at the magnetic compass which is equal and opposite to that produced at the same position by the degaussing coils.

6.2 Ordering data. Procurement documents should specify:

6.2.1 Procurement requirements.

(a) Title, number, and date of this specification.
(b) The required delivery dates for drawings, parts, and equipment.
(c) When test reports are required (see 4.6).
(d) Levels of preservation-packaging and packing required (see 5.1).

6.2.2 Contract data requirements. When this specification is used in a procurement invoking the data requirement clause of the Armed Services Procurement Regulations (ASPR) paragraph 7-104.9(n) and which incorporates a DD Form 1423 Contract Data Requirements List (CDRL), the data requirements identified below will be developed as specified in the cited Data Item Description (DID) and delivered in accordance with such CDRL. When the ASPR provisions are not invoked, the data specified below shall be delivered in accordance with the contract requirements.

Specification paragraph	Data requirements	Service	Applicable DID	Options
4.6	First Article Test Report	SH	UDI-T-23790	- - - -

(Copies of DID's required by the supplier in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.)

6.3 Government-furnished equipment. Procurement documents should specify Government-furnished equipment when furnished by the Government.

6.4 First article inspection.

6.4.1 Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection as to those bidders offering a product which has been previously procured or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending procurement.

6.5 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in Section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Preparing activity:
Navy - SH
(Project 5950-N165)

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL	OMB Approval No 22-R255

INSTRUCTIONS The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document DoD contractors, government activities or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government Fold on lines on reverse side, staple in corner, and send to preparing activity Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements Attach any pertinent data which may be of use in improving this document If there are additional papers, attach to form and place both in an envelope addressed to preparing activity

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3 IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO
	DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 68 WHICH MAY BE USED S/N 0102-014-1802

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T07:33Z
Check the source to verify that this is the current version before use.